

Aryan Shah

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EDUCATION

Virginia Tech

Bachelor of Science in Computer Science

Blacksburg, VA

Aug. 2023 – Dec. 2026

- **Relevant Coursework:** Advanced Data Structures & Algorithms, Computer Organization II, Combinatorics, Statistics

EXPERIENCE

VideoForce.ai (formerly CIPIO.ai)

AI Engineer Intern

Fairfax, VA

May 2026 – Aug. 2026

- Architected an end-to-end AI competitive-intelligence pipeline in **Python** (~6,000 LOC, 12 modules, 9 chained stages) that converts a single brand name into a full market-intelligence report on competitors' video ads, automating a research workflow that previously took analysts multiple days.
- Orchestrated Google Gemini on **GCP Vertex AI** across four multimodal tasks (competitor discovery, video speech transcription, creative classification, report generation) via a shared prompt layer and Cloud Storage staging service, enabling analysis of hundreds of video ads with zero manual review.
- Eliminated single-point-of-failure LLM outages by designing a shared resiliency layer that recovers from transient 503/429 errors via timed retry, prioritized model failover, and non-LLM fallbacks, allowing full runs to complete under provider rate limits.
- Cut ~13 minutes of redundant compute per run and removed duplicate API billing by parallelizing LLM transcription with per-item checkpointing and batching classification calls to Vertex AI's 10-video request cap, making cancelled runs resumable.
- Integrated three external **REST API** sources (Foreplay, SerpAPI, Playwright headless-browser fallback) and enforced enum-constrained model output plus a two-stage text- and vision-based relevance funnel, preventing hallucinated values from corrupting aggregate reporting.

Full Stack Engineer Intern

May 2025 – Aug. 2025

- Reduced manual email drafting time 98% by building an AI Reply Service integrating Groove webhooks, Google Gemini, and Supabase (**PostgreSQL**) into an event-driven backend workflow.
- Scaled the service to 50+ concurrent AI replies for a Fortune 500 beverage client while sustaining sub-100ms database operations, by introducing targeted indexing and asynchronous job queues.
- Increased agent throughput 10x and saved up to 5 minutes per email by shipping a **Next.js** review dashboard, and automated 100% of follow-ups with a CRON-scheduled SendGrid service.

Krossmark Innovations (Embee Group)

Software Engineering Intern

Gujarat, India

May 2024 – July 2024

- Improved inventory record-keeping efficiency 15% by building an online machine-inventory system with a team of 2 engineers using **Python**, **SQL**, and **React**.
- Cut manual task time 25% by designing full CRUD APIs for bill-of-materials management, then owning post-launch debugging and releases that improved system performance an additional 10%.

MachWorks @ Virginia Tech

Research & Development Intern

Blacksburg, VA

May 2024 – Sept. 2024

- Prototyped a supersonic AI-guided UAV and validated aerodynamic designs through **MATLAB** simulation and performance analysis, informing subsequent design iterations.
- Reduced project delays 25% and improved team efficiency 30% by producing CAD models and standardized technical documentation for a cross-functional engineering team.

PROJECTS

Edge AI Vision Assistant | *Python, Raspberry Pi, OpenAI API, Google TTS*

Jan. 2024 – May 2024

- Built a compact on-device AI assistant on Raspberry Pi that captures images and returns spoken answers in real time, chaining camera capture, an LLM inference call, and the Google Text-to-Speech API into one embedded Python pipeline.
- Achieved 95% speech-synthesis accuracy through iterative software optimization and latency tuning on constrained embedded hardware.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, TypeScript, JavaScript, HTML/CSS, MATLAB

AI/ML: LLM Orchestration, Prompt Engineering, Multimodal AI, Google Gemini, Vertex AI, Data Pipelines

Frameworks: React, Next.js, Node.js, Spring Boot, Playwright, FFmpeg, REST APIs, Async Job Queues

Cloud & Databases: Google Cloud Platform (Vertex AI, Cloud Storage), Supabase, PostgreSQL, MySQL

Tools & Certifications: Git, Docker, VS Code, Cursor, Twilio, SendGrid; IBM Python for Data Science, Stanford SQL